22 — Solid Track: DCHX Module & Bench Validation

v1.1 — coated-exchanger design, sizing, rig, and the M1–M4 measurement set

Function: Turn the doc 21 sorbent into a working thermal-swing module, and define the bench program that converts sizing-grade assumptions into measured numbers — above all **M2**, the single most economically consequential measurement in the program.

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1. Why a desiccant-coated heat exchanger (DCHX)

Coating the sorbent directly onto a finned/plate exchanger lets one element do both halves of the swing: **adsorb with cool fluid** in the channels (sink fluid removes the heat of adsorption — near-isothermal bed at full capacity); **regenerate with hot fluid** (reverse the loop; a sweep airstream carries vapor to the condenser train). DCHX outperforms packed granular beds (~90 vs ~59 kW/m^3; 3–5× heat-transfer rate) and is industrially proven in adsorption chillers — though packaged open-cycle DCHX *dehumidifiers* remain build-it-yourself, which is why this project exists. Geometry: flat-panel / stacked parallel plates first; higher-area-density geometries only if a plate stack cannot package.

Key sizing insight: throughput is governed by cycle time and coated area, not sorbent mass. Two (or more) modules alternate on ~10-minute half-cycles; short cycles with generous coated area beat a large slow bed. (Sensible-cycling caveat for the plumbing: doc 20 §4.)

2. Coating

Binder route (the only DIY-able route): sorbent + PVA slurry, sprayed (preferred over dip) at ~0.1–0.5 mm onto a prepared aluminum surface (acid-etched or boehmite-treated), ~0.18 kg sorbent/m^2 planform.

Binder rules (hard requirements):

- **PVA must be fully hydrolyzed (98–99%).** Partially hydrolyzed grades redissolve in humid service.
- **Even fully hydrolyzed PVA remains water-swellable near saturation unless heat-annealed.** The 120–150 °C activation **doubles as the mandatory anneal** — a re-coat must never skip it.
- Binder fraction low (~10 wt% nominal, locked by test).

Expected failure mode is **mechanical** (delamination / mass loss), not chemical — AlFu's framework is effectively bulletproof over 10^3–10^4 cycles. Durability testing targets the coating (M4).

3. Chloride exposure — finding F5 and the imported materials law

The DCHX is aluminum in a chloride-bearing environment (marine intake air; the liquid track's brine elsewhere on the platform; coastal land sites). Design responses, decided on paper:

- **Sealed/filtered intake path** — salt-aerosol filtration ahead of the coated face; in X8 closed-loop mode (doc 20 §8) the working path ingests no ambient aerosol at all, which is F5's primary mitigation.
- **No dissimilar-metal fittings** in the air path or fluid loop (extends the §5 bench rule to the installed module); no brass anywhere near the anodic plate.
- The sorbent + PVA coating is treated as a **barrier layer whose edge and defect behavior in salt air is inside M4's scope**.
- Imported from the liquid track's materials law (doc 11 §1): the **nylon/polyamide ban** near any $CaCl_2$ service, the **brazed-plate-HX prohibition** on chloride-wetted duty, drip-loop and containment discipline on any shared bulkheads, and the "pool/aquaculture parts, not plumbing parts" sourcing heuristic.

Corrosion *rate* is empirical and lands in M4; the responses above are not.

4. Sizing (annotated — do not use unannotated)

Baseline at 10-min half-cycles, two alternating modules, ~0.18 kg/m^2:

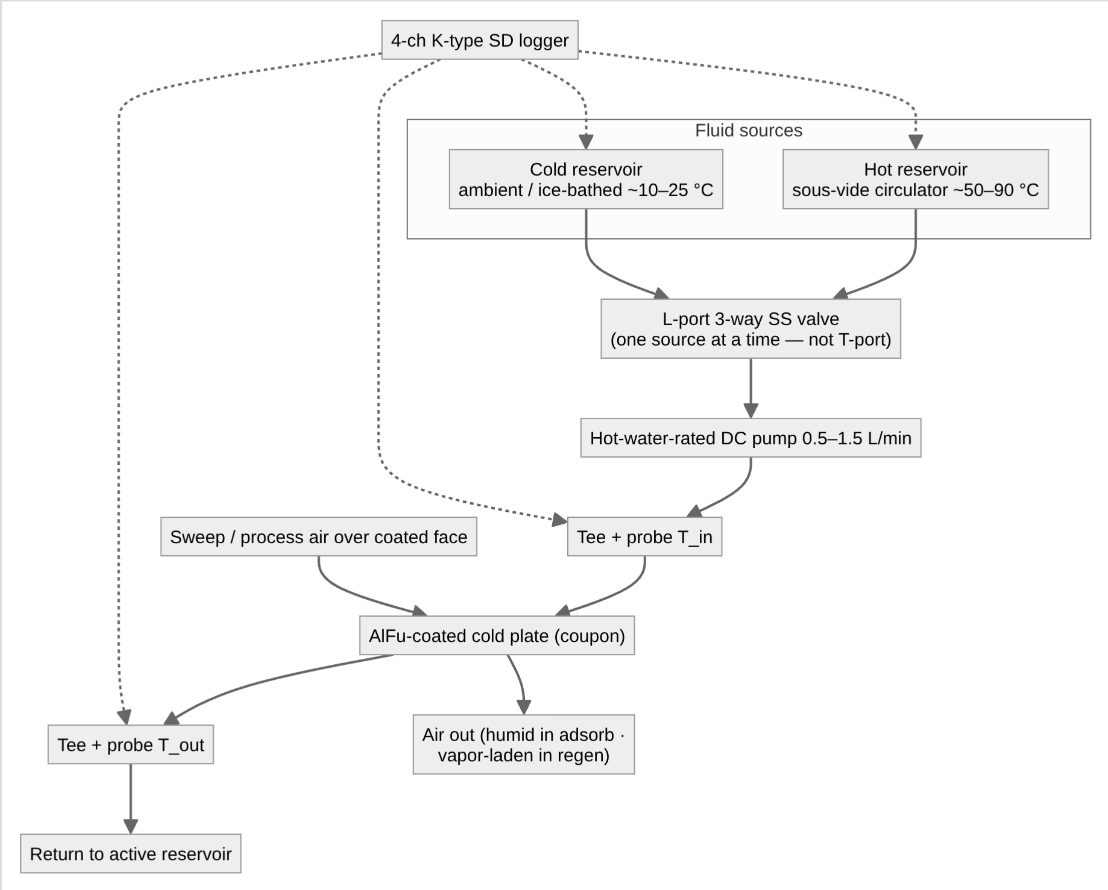
Basis	Duty	Δq basis	Sorbent inventory	Coated HX area
Envelope-only (naive; retained as the cautionary row)	~2–3 kg/h	0.2–0.3 g/g	~3–4 kg	~10–13 m^2
Self-consistent peak (doc 20 §5, PENDING T2)	**~9–11 kg/h**	0.2–0.3 g/g	~3–5× the above	scales accordingly
+ F3 compounded (PENDING M2)	~9–11 kg/h	effective ~0.15–0.2 g/g	+25–50% again	"

Open item — the coated-area denominator. The table above pairs ~3–4 kg of sorbent with ~10–13 m^2, which implies **~0.3 kg/m^2**, while §2 specifies **~0.18 kg/m^2 planform**. At 0.18 kg/m^2 the same inventory needs ~19 m^2. The likely explanation is a one-side-versus-two-side convention — a plate coated on both faces carries ~0.36 kg per m^2 of *plate* — but the documents do not say which denominator the area column counts, and the ambiguity scales into every derived area figure. Left flagged rather than resolved on paper: **M1 measures coated-face capacity and pressure drop on a known geometry** and fixes the denominator empirically. Until then, size area from the stated 0.18 kg/m^2 planform loading, which is the conservative reading.

The correction path — faster cycling vs more coated area vs architecture revision (warmer supply setpoint uses the comfort headroom to the 55% ceiling; hybrid brine pre-stage per doc 40 X-register) — is **open** and is exactly what the full T2 model decides. Any independent rebuild should treat T2 as the first modeling task, before hardware; the doc 00 §4 steady state is the interim basis.

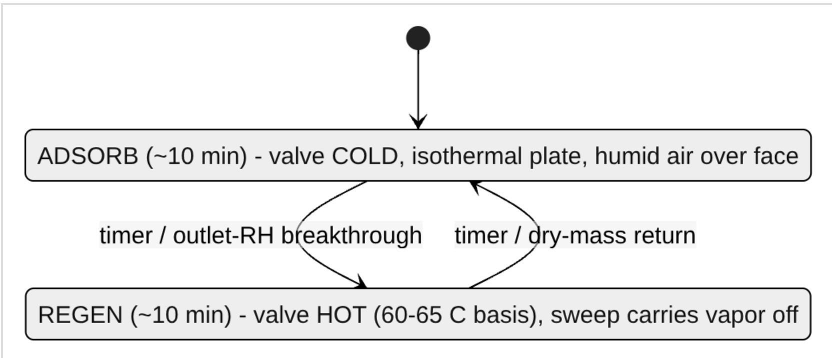
5. Bench test rig — single-coupon fluid loop

The coupon is a flat aluminum water-cooling block, AlFu-coated on the external face, temperature-controlled fluid in the channels. Switching the fluid source between hot and cold reservoirs is the whole trick; the fluid-side ΔT yields regeneration energy.



Loop rules: all-aluminum + stainless wetted path, distilled water (avoid brass — galvanic couple with the anodic Al plate); silicone tubing on the hot loop (PVC softens >60 °C); restrictive fittings on the pump *discharge* side; sous-vide circulator as the precision hot reservoir.

Duty cycle:



First characterization runs the phases *separately* (saturate for capacity, then clear for regeneration); cycling is for durability (M4).

6. The M1–M4 measurement set

ID	Measures	Method	Pass/insight
M1	Adsorption working capacity (g/g) **+ coated-face ΔP**	weigh activated-dry coupon → saturate isothermally at defined inlet RH → reweigh; **U-tube manometer across the coated face at design face velocity**	single-point capacity; ΔP gates the doc 20 fan-power line (0.6–1.0 kW claim)
M2	Regeneration vs **representative humid purge**	from saturated coupon, step hot setpoint across runs — one 45–50 °C confirmation (expected marginal per F2), then **completeness and kinetics at 60/65/70 °C** against a **logged ~24 g/kg purge, RH and ω recorded**	(a) desorption completeness/rate within the 10-min half-cycle at 60–65 °C; (b) **effective Δq under realistic purge** (the F3 number). Gates the heat-source decision
M3	Specific regeneration energy	fluid-side balance **$Q = \int \dot{m} \cdot c_p \cdot (T_{in} - T_{out})\, dt$**; dry/activated **blank** first (sensible + loss baseline, incl. the doc 20 §4 plumbing overhead), then wet; $(Q_{wet} - Q_{blank})/m_{water}$; condensate cross-check by weighing	vs ~0.67 kWh/L latent floor and ~0.8–1.0 kWh/L total
M4	Cycling durability **+ salt-air edge behavior**	repeat swing 10^2–10^3 cycles; periodic M1 + dry-mass weigh + visual; **coating edge/defect inspection under salt-aerosol exposure (F5)**	capacity fade and **coating mass loss/delamination** — the failure mode that matters

M2 is run against representative humid purge, never dry lab air. A dry-purge M2 reproduces the literature's ~50 °C claim and tells you nothing about the design point (doc 20 §6).

7. Instrumentation

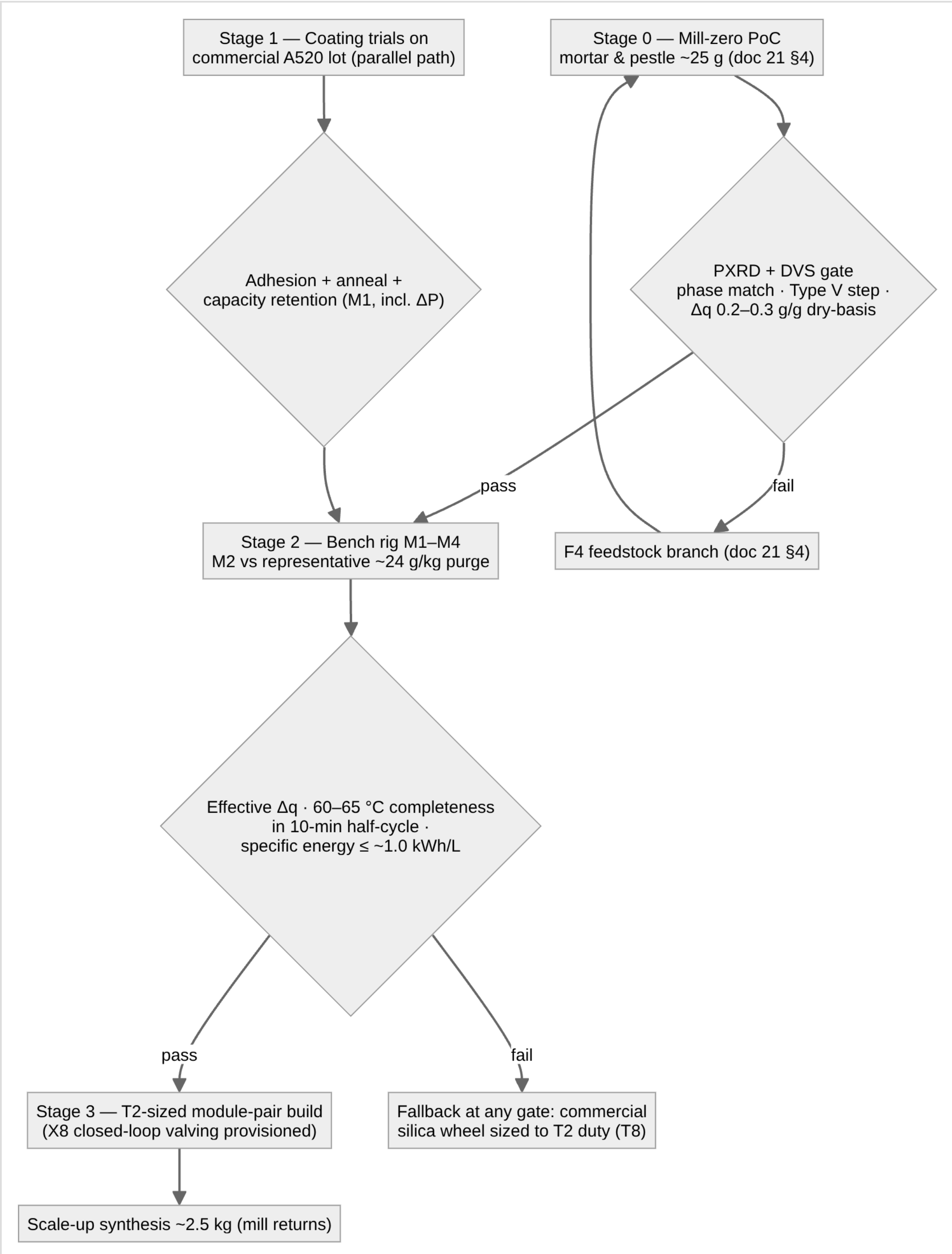
Fluid side: 4-channel K-type SD logger; four sealed SS 1/8″-NPT probes in inline tees — T1 plate inlet, T2 plate outlet (the ΔT pair), T3 hot, T4 cold. **Offset-calibrate the T1/T2 pair** (isothermal bath): an uncalibrated ±1 °C bias swamps a 3–5 °C signal in the M3 integral.

Air-side humidity — on the critical path, not optional:

Regime	Sensor	Why
Cool, sub-saturated nodes (post-desiccant)	digital capacitive RH/T (SHT85/SHT45 class)	accurate, fast; doubles as air-node thermometer
Near-saturated (>80% RH) and hot regen/purge (50–65 °C+) — the M2 nodes	**wet-bulb/dry-bulb psychrometry** (dry K-type + wicked aspirated wet K-type)	capacitive sensors are spec'd to ~60 °C / 20–80% RH, drift ~+3% RH in sustained saturation — exactly the regime that matters. This is also the instrument set tests D/G/H (doc 12) reuse

Wet/dry-bulb costs two K-type channels per air node; with the fluid loop consuming all four logger channels, plan the channel budget before M2 (second logger, or a unified ESP32 + MAX31856/SHT node with one timestamped CSV — the same ESP32 stack that runs the platform interlocks). Plus: a flow figure for the M3 integral (timed volumetric catch suffices) and a gravimetric balance. Oven setpoints independently verified by thermocouple — never trust a consumer oven dial for the activation/anneal step.

8. Staged validation pipeline



Version history

- **v1.1** — Clarifying pass from the parameter register (doc 50 §3.5), no design figure changed: §4 records the open coated-area denominator — the §4 sizing table implies ~0.3 kg/m² where §2 specifies 0.18 kg/m² planform — flagged against M1 rather than reconciled on paper, with the conservative reading named; the comfort headroom phrasing follows doc 20 v1.1's corrected 46% RH cabin state.
- **v1.0** — New lineage from archived 03: F5 chloride-exposure section and imported materials law added (§3); ΔP measurement added to M1; salt-air edge behavior added to M4; sizing table restated to the self-consistent duty; X8 valving provisioned at Stage 3.